

Sample Statistics

AGRON 5130

YOUR NAME HERE

Introduction

The goal of this assignment is to give you hands-on practice working with **sample statistics** and **confidence intervals**, building directly on the concepts introduced in this week's videos. You will work with real agronomic datasets, compute sample-based summary statistics, quantify uncertainty using the standard error, and construct confidence intervals using the t distribution.

Submit a knitted PDF or HTML generated from your R Markdown file to Canvas.

Question 1

1 point

Load the dataset "data/millet_uniformity.csv" and examine the first six rows using the `head()` function.

Pearl millet yield was sampled at 160 locations within a field in Kundewadi, Niphad, in the district of Nasik, India.

Question 2

1 points

Calculate the **mean**, **standard deviation**, and **number of observations** for millet yield.

Question 3

1 points

Calculate the **standard error of the mean** for millet yield.

Question 4

2 points

Calculate the t -value (using the `qt()` function) needed to construct a **95% confidence interval** for the mean yield.

A 95% confidence interval leaves 2.5% of the probability in each tail, so use 0.975. Remember that the degrees of freedom are one less than the number of observations.

Question 5

2 points

Calculate the **95% confidence interval** for millet yield using the statistics computed above.

Question 6

1 point

Load the dataset "data/safflower_uniformity.csv" and examine the first six rows using the `head()` function.

Safflower yield was sampled 340 times within a safflower field in Eloy, Arizona.

Question 7

1 points

Calculate the **mean, standard deviation, and number of observations** for safflower yield.

Question 8

2 points

Calculate the **standard error of the mean** for safflower yield.

Question 9

2 points

Calculate the t -value (using the `qt()` function) needed to construct a **90% confidence interval** for the mean safflower yield.

A 90% confidence interval leaves 5% of the probability in each tail, so use 0.95.

Question 10

2 points

Calculate the **90% confidence interval** for safflower yield using the statistics computed above.

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